Swarm Velocity Dynamics in US Sector ETFs *

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Abstract

This study utilizes the leader-following swarm velocity estimator of Chen et al. (2025) on eleven Select Sector SPDR ETFs at daily frequency and asks whether the estimated velocity carries tradable information. We backtest two strategies: a threshold rule that applies the cross-sectional average velocity to a leader-versus-loser pair trade, and a cross-sectional rule that ranks the ETFs by their own velocities. The swarm velocity proves to be informative but does not yield positive expected returns. The cross-sectional rule loses when trading from slow to fast ETFs but gains when doing the opposite, as the correlated ETFs cause the velocity estimator to favor an asset's own movement over its distance to the leader. Backtesting results reveal that all strategies are outperformed by an equally weighted portfolio of the same ETFs.

1 Introduction

A swarm is a population of simple agents whose collective motion emerges from local rules rather than central direction. Reynolds (1987) showed that the motion of flocks and schools emerges entirely from individual agents applying three local rules, namely separation, alignment and cohesion. Eberhart and Kennedy (1995) adapted the same idea into particle swarm optimization (PSO), in which each particle accelerates toward the best position it has found and toward the best position found by the swarm, so that the population converges on an optimum without gradients.

The financial interest lies in the leader term. Herding and lead-lag effects are documented features of asset returns, and momentum, which buys recent winners and sells recent losers, has been one of the most persistent anomalies since Jegadeesh and Titman (1993). A swarm model gives these phenomena a quantitative handle, since the coefficient on the leader term measures how quickly a follower closes the gap to the leader. Chen et al. (2025) estimate this speed for 190 risk factors at monthly frequency, sort the factors into quintiles, and report that slow-moving factors earn 0.26% per month more than fast-moving ones, a spread they attribute to delayed information processing.

This study extends the leader-following swarm velocity estimator of Chen et al. (2025) to eleven Select Sector SPDR ETFs at daily frequency. To assess whether the estimated velocity carries tradable information, we backtest a threshold rule that applies the cross-sectional average velocity to a leader-versus-loser pair trade and a cross-sectional rule that ranks the ETFs by their own velocities. Section 2 describes the data. Section 3 states the swarm velocity estimator. Section 4 specifies the threshold rule, and Section 5 specifies the cross-sectional rule. Section 6 concludes.

2 Data

This study focuses on the universe of eleven Select Sector SPDR ETFs. They together partition the S&P 500 into materials, communication services, energy, financials, industrials, technology, consumer staples, real estate, utilities, health care and consumer discretionary. Daily adjusted closing prices were collected from October 2018 to December 2025. The training sample runs from October 2018 to December 2023, and the testing sample runs from January 2024 to December 2025. Appendix B reports summary statistics.

*Code and replication data are available at *GitHub*.

3 Swarm Velocity

Each ETF is embedded in an L-dimensional space whose coordinates are its lagged returns, and the empirical velocity is the change in position between consecutive days,

$$\boldsymbol{x}_{i,t} = (r_{i,t-1}, \ldots, r_{i,t-L}), \qquad \boldsymbol{\nu}_{i,t} = \boldsymbol{x}_{i,t} - \boldsymbol{x}_{i,t-1}, \qquad \boldsymbol{d}_{i,t} = \boldsymbol{x}_{\ell_t,t} - \boldsymbol{x}_{i,t}, \tag{1}$$

with $L = 5$ and ℓ_t the leader. The leader-following model states that velocity is proportional to the gap, $\boldsymbol{\nu}_{i,t} = \alpha \boldsymbol{d}_{i,t}$, and the least squares estimate for a single ETF, together with the estimate obtained by imposing one common speed on the whole swarm, is

$$\hat{\alpha}_t^{(i)} = \frac{\langle \boldsymbol{\nu}_{i,t}, \boldsymbol{d}_{i,t} \rangle}{\|\boldsymbol{d}_{i,t}\|^2}, \qquad \hat{\alpha}_t = \frac{\sum_i \langle \boldsymbol{\nu}_{i,t}, \boldsymbol{d}_{i,t} \rangle}{\sum_i \|\boldsymbol{d}_{i,t}\|^2}. \tag{2}$$

Appendix A includes the derivation and shows that $\hat{\alpha}_t$ is the precision-weighted combination of $\{\hat{\alpha}_t^{(i)}\}$. The leader is identified as the ETF with the highest cumulative return over the position window. It satisfies $\boldsymbol{d}_{\ell_t,t} = \mathbf{0}$ and is excluded.

On the training sample $\hat{\alpha}_t^{(i)}$ has mean -0.388 with standard deviation 0.814 and $\hat{\alpha}_t$ has mean -0.468. Both are negative, whereas Chen et al. (2025) report that every estimated speed is positive, with a mode slightly below 0.5. This stems from the estimator's composition, where the eleven sectors have a mean pairwise return correlation of 0.65. Since the numerator $\langle \boldsymbol{\nu}, \boldsymbol{d} \rangle$ consists of a pull toward the leader and the asset's own increment, its own increment dominates when a sector sits close to the leader. The 190 heterogeneous factors at monthly frequency in Chen et al. (2025) have a weaker correlation, so the pull toward the leader dominates and the estimate is positive.

4 Threshold Rule

To investigate how the swarm provides a quantitative approach to momentum strategies, we backtest a simple threshold-based trading strategy. The threshold rule trades a pair between the day's leader $\ell_t = \arg\max_i r_{i,t}$ and the day's loser $s_t = \arg\min_i r_{i,t}$, and uses $\hat{\alpha}_t$ to decide whether to trade at all. With $\boldsymbol{e}_i$ the ith unit vector and thresholds $\tau_l \le \tau_u$,

$$\boldsymbol{w}_t = \tfrac{1}{2} \begin{cases} +\boldsymbol{e}_{\ell_t} - \boldsymbol{e}_{s_t}, & \hat{\alpha}_t > \tau_u \text{ and } r_{\ell_t,t} > 0, \\ -\boldsymbol{e}_{\ell_t} + \boldsymbol{e}_{s_t}, & \hat{\alpha}_t < \tau_l \text{ and } r_{\ell_t,t} < 0, \\ \mathbf{0} & \text{otherwise}, \end{cases} \tag{3}$$

Half the capital is long, and half is short, with no leverage. The evaluation is repeated on a daily basis. Thresholds are optimized on the training sample over the grid $\tau_l \in \{-1.3, -1.2, \ldots, -0.4\}$ and $\tau_u \in \{-0.4, -0.3, \ldots, 1.5\}$.

To isolate the contribution of the velocity metric, the benchmark is the same pair trade without the velocity condition. It buys the leader and sells the loser when the leader's return is positive, and sells the leader and buys the loser when it is negative. The backtest is conducted under a transaction cost of five basis points per unit of one-way turnover.

The backtest results demonstrate that the velocity metric is informative but does not make the rule profitable. The first two rows of Table 1 compare the rule with its benchmark. The benchmark loses 67.3% of capital in training and 45.8% in testing. It holds a position 99.8% of the time, and frequent turnover erodes 21.8% of capital in transaction costs. The velocity metric with a threshold reduces the number of trades that lose or make little profit. The strategy under the threshold rule is active on 10.4% of training days and 4.8% of testing days. It yields a +1.58% training outcome and cuts the annual cost to 2.4% of capital. In this narrow sense, the velocity metric is informative because it identifies which trades not to take.

However, the backtest results show that it does not add a positive expectation. The strategy returns +1.58% in training at a Sharpe ratio of +0.083, but −7.29% in testing at −1.19. Figure 1 shows that this is not a matter of threshold choice. 9 of 200 training cells and 2 of the testing cells are profitable. The best cell in the testing surface returns +0.2%, and most thresholds makes this trading strategy unprofitable.

	Training			Testing		
	Return	Volatility	Sharpe	Return	Volatility	Sharpe
Threshold rule at (−0.5, 0.3)	+1.58%	5.36%	+0.083	−7.29%	3.18%	−1.19
Benchmark	−67.3%	14.9%	−1.36	−45.8%	11.1%	−2.74
Opposite rule at the same pair	−24.5%	5.40%	−0.971	+3.01%	3.17%	+0.492
Opposite benchmark	−72.2%	14.9%	−1.57	−24.7%	11.1%	−1.24

Table 1: Backtest performance table; volatility is annualized.

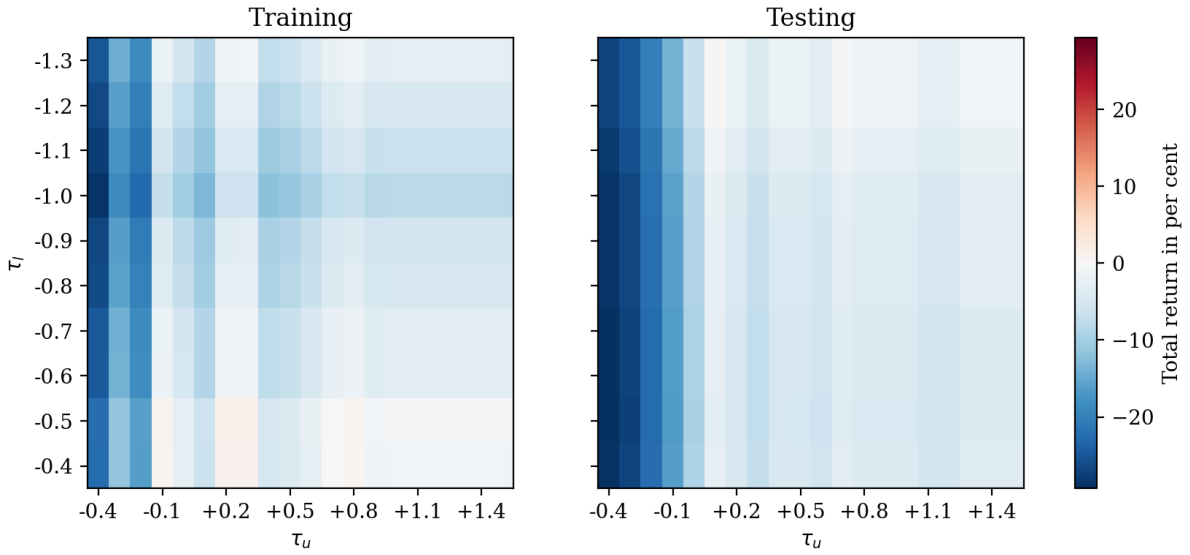


Figure 1: Total return of the threshold rule over the grid of thresholds.

Since the trade loses under almost every calibration, we naturally ask whether its direction is simply inverted. We reverse the long-short position and the results are shown in the lower block of Table 1. The answer is negative. Reversing the benchmark shows no improvement in training at −72.2%. The opposite rule loses 24.5% in training, where the main rule gains, and gains 3.01% in testing, where the main rule loses.

Therefore, the empirical findings suggest that the swarm velocity metric provides a quantitative approach that can help identify which trades not to take. The unprofitable backtest result is attributed to the trivial trading rule. This class of momentum strategy is highly sensitive to market conditions, and that sensitivity is a property of the underlying leader-versus-loser trade, not the velocity metric.

5 Cross Sectional Rule

The cross-sectional rule follows the design of Chen et al. (2025). On each formation day, the ETFs are ranked on $\hat{\alpha}_t^{(i)}$, the slow minus fast rule refers to the k slowest being bought and the

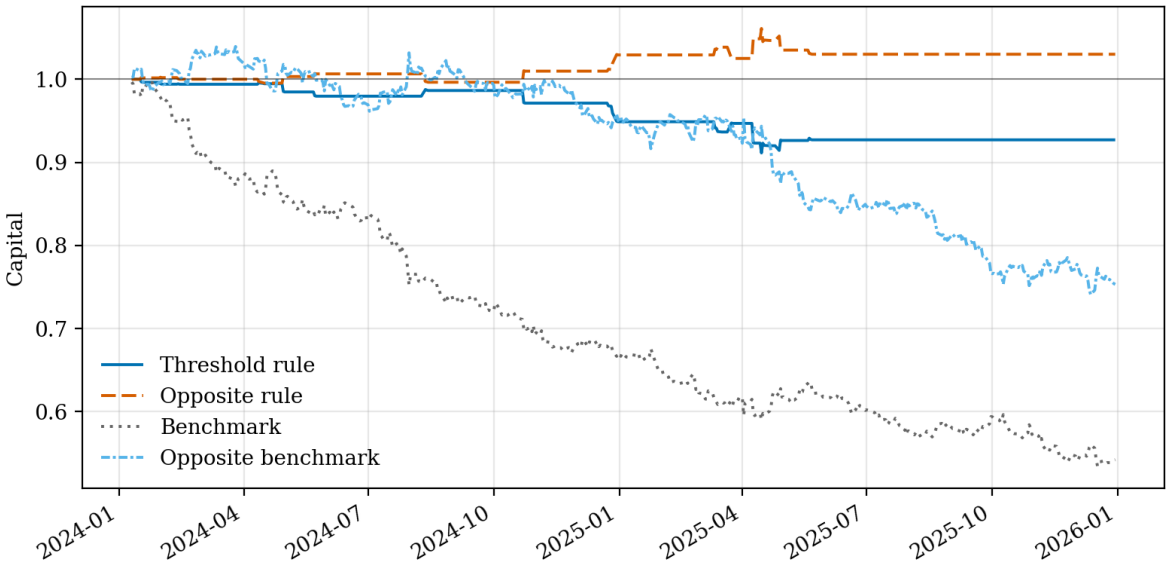


Figure 2: Capital of the threshold rule.

k fastest being sold in equal weight,

$$\boldsymbol{w}_t^{(1)} = \frac{1}{2k}\left(\sum_{i\in\mathcal{L}_t} \boldsymbol{e}_i - \sum_{i\in\mathcal{H}_t} \boldsymbol{e}_i\right), \tag{4}$$

Conversely, the fast minus slow rule refers to the k slowest being sold and the k fastest being bought in equal weight,

$$\boldsymbol{w}_t^{(2)} = \frac{1}{2k}\left(\sum_{i\in\mathcal{H}_t} \boldsymbol{e}_i - \sum_{i\in\mathcal{L}_t} \boldsymbol{e}_i\right), \tag{5}$$

where $\mathcal{L}_t$ and $\mathcal{H}_t$ collect the k lowest and k highest estimates, so that half the capital is long and half is short. We set $k = 2$, reproducing the breadth of one-fifth on each side. The evaluation is repeated every h trading days, and the baseline is set to a one-month horizon, $h = 21$.

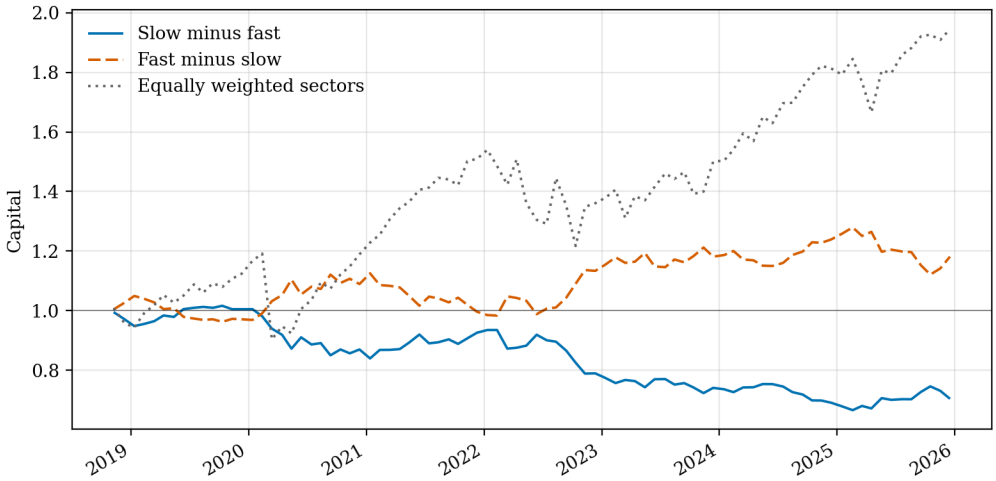


Figure 3: Capital of the cross-sectional rule, against an equally weighted portfolio.

Figure 3 shows the backtest performance. The slow-minus-fast strategy returns -29.6% over the sample at a Sharpe ratio of -0.56, and the fast-minus-slow strategy returns $+18.1\%$ at a

Sharpe ratio of +0.33. We benchmark these strategies with an equally weighted portfolio of the eleven ETFs rebalanced on the same dates. The equally weighted benchmark returns 94.5% at a Sharpe ratio of +0.65, completely dominating both strategies.

To test the robustness of the holding period h, Table 2 and Figure 4 vary h around that baseline over $\{1, 2, 3, 5, 10, 21, 63\}$. The slow-minus-fast strategy has negative performance for all holding periods, while the fast-minus-slow strategy yields positive returns when h is at least 10. On the one hand, shorter h implies higher total turnover, which increases transaction costs. More importantly, the information in the sort accumulates with the horizon. The mean cross-sectional rank correlation between $\hat{\alpha}_t^{(i)}$ and the subsequent return rises from +0.008 at 5 days to +0.016 at 10 days, +0.018 at 21 days and +0.071 at 63 days. Both of these factors support the idea of a longer holding period.

	Slow minus fast		Fast minus slow	
h	Return	Sharpe	Return	Sharpe
1	−70.6%	−2.16	−49.7%	−1.19
2	−63.4%	−1.70	−17.7%	−0.30
3	−53.1%	−1.34	−14.0%	−0.24
5	−39.5%	−0.84	−12.6%	−0.20
10	−35.2%	−0.72	+11.7%	+0.23
21	−29.6%	−0.56	+18.1%	+0.33
63	−21.9%	−0.44	+18.0%	+0.36

Table 2: Total return and Sharpe ratio of the cross-sectional rule by holding period h in trading days. Annualized volatility lies between 7.4% and 8.1% in every cell.

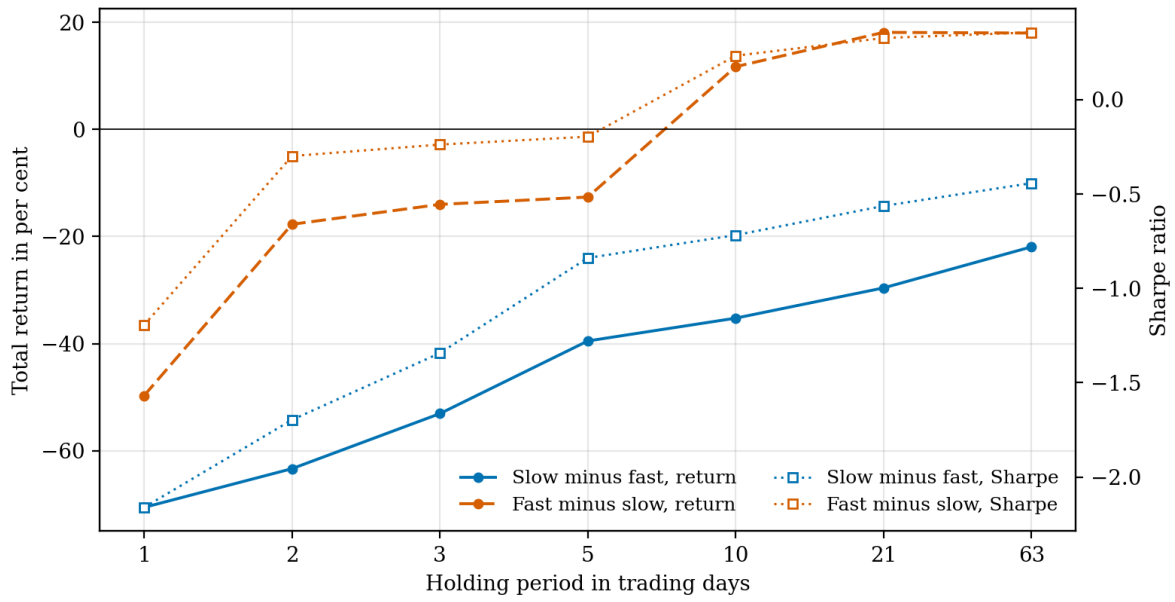


Figure 4: Total return and Sharpe ratio of the cross-sectional rule against the holding period.

It should be noted that the sign of that rank correlation is positive, which differs from what Chen et al. (2025) report for 190 factors. In the universe of eleven ETFs, the fast movers earn higher subsequent returns, and the profitable portfolio is long fast and short slow. The difference lies in the composition of the estimator described in Section 3. When the increment outweighs the pull, a high estimate indicates an ETF price has significantly shifted in the last day, rather

than one that quickly closes in on the leader. In this case, one should not expect the direction of its subsequent return to align with the direction observed when the pull is dominant.

6 Conclusion

The swarm velocity estimator transfers to eleven sector ETFs as a quantitative approach but not as a tradable signal. The sign of the velocity estimate inverts relative to Chen et al. (2025) because pairwise return correlation keeps sectors clustered near the leader. The velocity calculation is dominated by an asset's own recent price movement rather than its structural speed in closing the distance to the leader.

Across both backtested strategies, the empirical results show clear constraints. Incorporating a velocity threshold improves performance by filtering out unprofitable trades and reducing turnover costs. However, it fails to yield positive out-of-sample expected returns. Sorting assets by velocity shows positive returns when trading from fast to slow ETFs, particularly over longer time horizons. With a baseline monthly holding period, this strategy returns 18.1% at a Sharpe ratio of +0.33, but substantially underperforms an equally weighted portfolio benchmark, which posts a return of 94.5% at a Sharpe ratio of +0.65.

Ultimately, these findings demonstrate that turnover dynamics and transaction costs largely govern strategy performance over a narrow, correlated universe of eleven sector ETFs. While swarm velocity provides information for trade filtering, it does not offer a standalone tradable edge.

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A Closed Form for the Velocity Estimate

Imposing one common speed on the swarm, the sum of squared errors at date t is

$$S(\alpha) = \sum_i \|\boldsymbol{\nu}_{i,t} - \alpha \boldsymbol{d}_{i,t}\|^2$$

$$S'(\alpha) = -2\sum_i \langle \boldsymbol{\nu}_{i,t}, \boldsymbol{d}_{i,t} \rangle + 2\alpha \sum_i \|\boldsymbol{d}_{i,t}\|^2, \qquad S''(\alpha) = 2\sum_i \|\boldsymbol{d}_{i,t}\|^2 > 0,$$

so S is strictly convex whenever at least one gap is non-zero, and its unique minimizer is the second expression in (2). Restricting the sum to a single term gives the first. The two are related by

$$\hat{\alpha}_t = \sum_i \frac{\|\boldsymbol{d}_{i,t}\|^2}{\sum_j \|\boldsymbol{d}_{j,t}\|^2} \, \hat{\alpha}_t^{(i)},$$

so the pooled estimate weights each ETF by the squared length of its own gap.

B Summary Statistics

ETF	Annualized return (%)	Annualized volatility (%)	Skewness	Excess kurtosis
XLB	8.1	22.5	-0.41	9.6
XLC	13.1	22.9	-0.48	7.1
XLE	6.6	33.1	-0.94	14.8
XLF	11.4	24.2	-0.58	15.1
XLI	11.0	21.9	-0.51	13.0
XLK	19.4	27.0	-0.27	8.3
XLP	7.8	15.9	-0.46	15.5
XLRE	6.4	22.7	-1.04	17.1
XLU	9.9	21.0	-0.25	15.1
XLV	8.4	17.8	-0.40	10.1
XLY	10.8	24.7	-0.53	7.4

Table 3: Summary statistics, annualized. Every sector is left skewed and heavy tailed.

	XLB	XLC	XLE	XLF	XLI	XLK	XLP	XLRE	XLU	XLV	XLY
XLB	1.00										
XLC	0.68	1.00									
XLE	0.67	0.45	1.00								
XLF	0.84	0.68	0.71	1.00							
XLI	0.89	0.70	0.69	0.89	1.00						
XLK	0.70	0.83	0.45	0.67	0.75	1.00					
XLP	0.66	0.56	0.42	0.65	0.66	0.54	1.00				
XLRE	0.71	0.60	0.48	0.69	0.72	0.59	0.71	1.00			
XLU	0.59	0.45	0.40	0.58	0.60	0.45	0.73	0.76	1.00		
XLV	0.71	0.64	0.47	0.69	0.72	0.66	0.74	0.69	0.63	1.00	
XLY	0.75	0.82	0.49	0.73	0.78	0.84	0.57	0.65	0.46	0.63	1.00

Table 4: Correlation matrix. The mean off-diagonal entry is 0.65, and the range is 0.40 to 0.89.